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EVEN END SEMESTER EXAMINATION – 2024

Name of the Program: B.Tech.

Semester: III

Name of the Course: Introduction to Cryptography

Paper Code: TCS-392

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks for each main question are twenty.
- (iv) Each sub question carries 10 marks

Q1	(20marks)	CO1
(a)	Explain security architecture. Justify its relevance and significance in providing security.	
(b)	Which security services are guaranteed when using each of the following methods to send packages via the post office? i. Standard package service ii. Package with tracking iii. Package with insurance and signature confirmation	
(c)	Classify the various security attacks involved in data communication over the Internet. Illustrate your answer with a diagram showing the taxonomy of these attacks.	
Q2	(20 marks)	CO2
(a)	(i) Alice wants to send a secret message "I Love My Country" to bob. She decided to use the Ceaser cipher to encrypt the message with key value 4. Help Alice to encrypt the message. (ii) Detective Vijay intercepted a message "WTAAD", and after analysis he found that it has been encrypted using Caesar cipher and probable key is 15. Help the Vijay to decipher the message.	
(b)	Use Playfair cipher to encrypt the following messages with key "Computer" (i) Balloon (ii) Mosque	
(c)	(i) Encrypt the message "This is a secret message" using Rail fence Cipher with key 3. (ii) Describe the key components of the Data Encryption Standard (DES) algorithm, also explain its encryption process.	
Q3	(20 marks)	CO3
(a)	Analyze and explain the steps involved in key distribution between two parties A and B via KDC. Where A and B both share a secret key with KDC.	
(b)	(i) Explain Euclid's algorithm for finding GCD of two positive integers. (ii) Discuss the importance of random numbers in cryptography. Additionally, differentiate between true random numbers and pseudo-random numbers.	
(c)	Discuss hierarchical key control and session key lifetime in context of key distribution.	
Q4	(20 marks)	CO4
	In an RSA cryptosystem, a particular user A uses two prime numbers $p = 3$ and	

(a)	q = 11 to generate her public and private keys. If the public key of A is (7, n) and plaintext M is 31. Then find the private key of A and cipher text.	
(b)	Discuss the need of public key cryptography along with the principal elements of public-key cryptosystem.	
(c)	Classify and provide details of the different types of malicious software with neat figure.	CO5
Q5	(20 marks)	
(a)	List and briefly define classes of intruders. Also compare statistical anomaly detection and rule-based intrusion detection system.	CO5
(b)	(i) Identify and Discuss computer crime, based on the role that the computer plays in the criminal activities. (ii) What do you think are the challenges faced by Law enforcement agencies.	CO6 CO6
(c)	Identify three main types of Intellectual properties where legal protection is available. Elaborate each of the types in detail.	